

Orest Kupyn

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Computer Vision Researcher • 4,000+ Citations • 10+ Years Scaling ML from Research to Production • Oxford VGG & Google Research

Summary

PhD researcher at the Visual Geometry Group (Oxford) and Google Research (ELLIS Industry PhD) specializing in generative models, synthetic data generation, and 3D/video generation. Combined frontier research (4,000+ citations, published at top-tier venues, 2.6k+ open-source tools) with 10+ years of engineering experience including founding a regional ML office, building research teams from scratch, and deploying computer vision systems serving millions of users at production scale.

Education

- University of Oxford — Visual Geometry Group** Oxford, United Kingdom
DPhil in Engineering Science (ELLIS Industry PhD with Google Research) Oct 2023 — Present
– Research on generative models for geometric video generation, synthetic data, and 3D scene generation. Supervised by Prof. Christian Rupprecht. Industrial co-supervision at Google Research by Dr. Federico Tombari
- Ukrainian Catholic University** Lviv, Ukraine
M.Sc. in Computer Science — Data Science Specialization 2016 — 2017
- Lviv National University** Lviv, Ukraine
B.Sc. in Computer Science — Information Systems 2012 — 2016

Industry Experience

- Google Research** Zurich, Switzerland
ELLIS Industry PhD Researcher Jan 2026 — Present
– **Scene Generation:** Research on simulation-ready, geometrically accurate 3D scene generation.
- PiñataFarms** (backed by Index Ventures & Founders Fund) Lviv, Ukraine
Founding Engineer & Research Lead May 2019 — Oct 2023
– **Founding-Stage Leadership:** Second engineer at the company. Set up and opened the company's regional engineering office in Ukraine. Defined technical vision and architecture for all ML systems. Grew the office from 2 to 50+ employees.
– **ML Research Team:** Built and led 10-person ML research team from scratch. Established in-house data labeling pipeline and acquisition capabilities. Research initiatives resulted in publications at top-tier conferences (CVPR, ECCV) while simultaneously serving production traffic.
– **Production Computer Vision Platform:** Architected end-to-end real-time computer vision pipeline: 3D Head Reconstruction, Head Segmentation, Real-Time Object Tracking, Interactive Segmentation, and content recommendation. Served millions of monthly active users.
- SoftServe** Lviv, Ukraine
Senior Research Engineer Aug 2018 — May 2019
– **Google Partnership:** Led development of seismic data segmentation and horizon detection module as subcontractor for Google Oil and Gas Platform.
- WANNABY** (Acquired by FarFetch) Minsk, Belarus
Research Engineer Mar 2018 — Aug 2018
– **AR Computer Vision:** Developed GAN-based texture recoloring models for AR try-on applications and 3D foot reconstruction algorithms deployed in consumer mobile apps.
- Eleks, SoftServe, Vakoms** Lviv, Ukraine
Software Engineer May 2013 — Apr 2018
– **ML & Mobile Applications:** Satellite imagery segmentation for agriculture, text analysis platforms for manufacturing, iOS development for enterprise videoconferencing.

Selected Publications

- MMDiff: Extending Diffusion Transformers for Multi-Modal Generation.** Y. Akarken, O. Kupyn, C. Rupprecht. *Under review, 2026.*
- VGGRPO: Towards World-Consistent Video Generation with 4D Latent Reward.** Z. An, O. Kupyn, T. Uscidda, A. Colaco, K. Ahuja, S. Belongie, M. Gonzalez-Franco, M. Tintore Gazulla. *Under review, 2026.*
- PoseDreamer: Scalable Photorealistic Human Data Generation with Diffusion Models.** L. Prospero, O. Kupyn, O. Viniavskiy, J. F. Henriques, C. Rupprecht. *Under review, 2026.*
- Epipolar Geometry Improves Video Generation Models.** O. Kupyn, F. Manhardt, F. Tombari, C. Rupprecht. *Under review, 2025.*

- **S3OD: Towards Generalizable Salient Object Detection with Synthetic Data.** O. Kupyn, H. Kataoka, C. Rupprecht et al.. *ICLR 2026*.
- **VGGHeads: A Large-Scale Synthetic Dataset for 3D Human Heads.** O. Kupyn, E. Khvedchenia, C. Rupprecht. *CVPRW 2026*.
- **Dataset Enhancement with Instance-Level Augmentations.** O. Kupyn, C. Rupprecht. *ECCV 2024* — **Oral Presentation**.
- **FEAR: Fast, Efficient, Accurate and Robust Visual Tracker.** V. Borsuk, R. Vei, O. Kupyn, T. Martyniuk, I. Krashenyi, J. Matas. *ECCV 2022*.
- **DAD-3DHeads: A Large-Scale Dense, Accurate and Diverse Dataset for 3D Head Alignment.** T. Martyniuk, O. Kupyn, Y. Kuriyak, I. Krashenyi, J. Matas, V. Sharmanska. *CVPR 2022*.
- **DeblurGAN-v2: Deblurring (Orders-of-Magnitude) Faster and Better.** O. Kupyn, T. Martyniuk, J. Wu, Z. Wang. *ICCV 2019*.
- **DeblurGAN: Blind Motion Deblurring Using Conditional Adversarial Networks.** O. Kupyn, V. Budzan, M. Mykhailych, D. Mishkin, J. Matas. *CVPR 2018*.

Open Source & Impact

DeblurGAN — 2,600+ GitHub stars, Arctic Code Vault contributor, widely adopted in production. **DAD-3DHeads** — 470+ stars, large-scale dataset and benchmark for 3D head alignment (CVPR 2022). **VGGHeads** — 200+ stars, synthetic dataset for 3D head alignment. **S3OD** — Open-source synthetic data generation pipeline for salient object detection. Total: **4,000+ Google Scholar citations**.

Technical Skills

Research: Diffusion Models, Video Generation, 3D Reconstruction, DPO/RLHF Alignment, Flow Matching, GANs, Synthetic Data Generation
Engineering: PyTorch, Large-Scale Distributed Training, Real-Time Inference Systems, Production ML Pipelines, Python, C++
Leadership: Founded regional engineering office; built and managed 10-person research team; scaled organization from 2 to 50+

Patents

- **US12361710B2** — System for remixable video content and per-frame metadata capture and playback. C. Carrier, C. Adame, L. Schulte, **O. Kupyn**, A. Hakeem. PiñataFarms Inc., granted 2025.

Honors & Awards

- **Outstanding Reviewer** — CVPR 2026
- **Samsung AI Challenge** — 2nd Place, Samsung Research AI Hackathon for neural image restoration (2018)
- **Queen's Innovation Challenge** — Winner, CAD \$20,000 prize for AI plant disease recognition (2017)
- **Data Science Game** — International Finalist, ML & Data Science competition, Paris (2017)